Data-Driven Battery Replacement Analytics: Penske and Saint Joseph's University Partnership Report

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1 EXECUTIVE SUMMARY

This report summarizes a collaborative analytics initiative between Penske Truck Leasing and Saint Joseph's University. This project aimed at improving battery maintenance through predictive analytics. Using advanced statistical and machine learning methods, the project focused on predicting battery replacements, forecasting repair times, detecting anomalous replacement events, and extracting insights from maintenance job notes.

We developed a highly accurate binary classification model using the artificial intelligence algorithm-Random Forest to predict the battery replacements, achieving exceptional predictive performance (Accuracy: 99.13%, Precision: 99.52%, Recall: 99.53%, F1-score: 99.53%). The analysis highlighted mileage as the most critical predictor, emphasizing the significant role of cumulative distance in assessing battery health and risk. Other influential factors included detailed odometer percentiles, duty cycle metrics, cranking voltage variability, and operational indicators such as vehicle configuration flags (e.g., liftgate, sleeper, reefer). To enhance model interpretability, we utilized LIME—an advanced tool designed to explain complex, black-box AI models. LIME provided clear explanations for individual predictions, further validating our key findings. Notably, it confirmed that high mileage (especially for trucks with mileage exceeding 343,021 miles) and specific telematics metrics, such as vehicle configuration flags, are strong predictors of battery replacements.

To enhance operational planning, we applied LASSO regression to predict the time between telematics alerts and battery repairs, achieving an average prediction error of 17 days. Features such as minimum odometer readings, lower-end ambient temperatures, and certain cranking voltage percentiles correlated with longer time to repair, potentially due to less obvious or urgent issues. Conversely, features like maximum odometer readings, higher cranking voltages, and specific vehicle configurations (e.g., sleeper or liftgate equipped) correlated with shorter intervals, likely reflecting more urgent or critical conditions. These insights enable better maintenance scheduling, quicker responses, and improved operational efficiency.

We employed a One-Class SVM, an advanced unsupervised learning approach capable of independently identifying abnormal telematics patterns within the battery replacement dataset. Unlike traditional supervised methods, this model learns the dominant, typical patterns directly from telematics data, enabling it to effectively flag anomalies without prior labeled examples. Our analysis identified anomalous battery replacements characterized by significantly elevated odometer readings and declining cranking voltages several weeks before replacement. Key telematics indicators—such as the 75th percentile of cranking voltage eight weeks prior and odometer readings five weeks prior—were particularly influential in distinguishing outliers from typical replacement cases.

Text mining of unstructured technician job notes revealed recurring operational themes such as systematic battery diagnostics, alternator and electrical system issues, and emergency jump-start scenarios. To extract deeper insights from technician job notes, we applied Latent

Dirichlet Allocation (LDA), an advanced topic modeling technique that uncovered five key operational themes within the unstructured text data. The identified topics and representative keywords were: (1) Battery Testing and Replacement Procedures (battery, test, voltage, replaced, cable); (2) Truck Battery Jump Start and Service (unit, jump, start, truck, shop, customer); (3) Alternator and Electrical System Diagnostics (alternator, system, belt, electrical, test, replace); (4) Battery Inspection and Reassembly Process (check, inspect, remove, reassemble, verify); and (5) Battery and APU System Maintenance (battery, replace, test, check, vendor, APU, outside).

In summary, the integration of predictive analytics significantly advances Penske's fleet reliability and operational efficiency. Next steps include deploying predictive alerts, refining maintenance protocols based on analytics-driven insights, and continuously updating models to maintain high predictive accuracy.

2 INTRODUCTION

2.1 PROJECT BACKGROUND

Penske Truck Leasing, a leading provider of transportation solutions, operates and maintains a fleet of over 437,000 vehicles across North America. Ensuring the reliability and efficiency of this extensive fleet is paramount, as unexpected breakdowns can lead to significant operational disruptions and increased maintenance costs.

Traditionally, maintenance schedules have been based on fixed intervals, which may result in unnecessary replacements or overlooked issues. To enhance maintenance strategies, in this project, we are leveraging predictive analytics to transition from reactive to proactive maintenance approaches. By analyzing data from battery maintenance records, telematics systems, and engine system ratings, predictive models can identify early signs of potential battery failures, predicts whether a battery replacement is necessary or unnecessary. This enables timely interventions, reducing unexpected breakdowns and extending battery life.

2.2 OBJECTIVES

This report unifies four complementary objectives: predicting which vehicles need electrical system diagnosis, forecasting how many days remain before an electrical system failure, detecting anomaly battery replacement events and extracting recurring insights from technicians' notes. To be specific:

- Binary Classification Model for Electrical System Failure: Develop a predictive model to classify if the electrical system needs to be diagnosed. This aims to optimize maintenance decisions, reduce unnecessary replacements, and minimize vehicle downtime, thereby enhancing operational efficiency and cost-effectiveness.

- Time-to-Repair Prediction Using LASSO Regression: Utilize LASSO regression to forecast the expected duration of battery repairs. Accurate predictions will assist in resource planning, improve scheduling efficiency, and reduce service disruptions, contributing to better customer satisfaction and operational planning.
- Outlier Detection in Battery Repair Data: Deploy a One-Class SVM to learn the "normal" multivariate pattern in the battery-replacement telematics data, automatically flagging anomalous replacement events that deviate from this baseline. After isolating these anomalies, we will rank-order the telematicsthat most strongly contribute to the outlier score, using SHAP or permutation-importance analysis.
- Text Mining on Job Notes: Apply text mining techniques to analyze unstructured job notes from technicians. This will extract valuable insights into recurring issues and maintenance trends, informing training programs, enhancing maintenance protocols, and ultimately improving service quality

2.3 DATA SOURCE AND OVERVIEW

Here are several datasets we used for this project:

1. Telematics Hourly Data: A folder containing 160 Parquet files with hourly telematics data collected from January 1 to December 30, 2024. Each file represents a specific time segment and captures various operational and telematics metrics at an hourly frequency. After combining all files, the final dataset includes 1,048,576 rows, 38 columns, and 1,024 unique units.
2. Telematics Weekly Data: A folder containing 200 Parquet files with weekly telematics data collected from December 26, 2022 to November 18, 2024. Each file captures various operational and telematics metrics at an hourly frequency. After combining all files, the final dataset includes 527,180 rows, 38 columns, and 14,006 unique units.
3. Battery Repair Data: The dataset comprises 299,512 rows and 28 columns, capturing detailed maintenance and operational records for various vehicle units. Key variables include identifiers such as unit_num, category, make_code, and model_year, which provide basic information about each vehicle. Engine-related attributes such as engine_make, engine_model, engine_model_year, unit_engine_epa_year, and unit_engine_epa_type offer insights into engine specifications and compliance data. Maintenance and repair activities are tracked through fields like workorder, wo_start_date, wo_complete_date, job_start, job_complete, and profile_job_description, providing a timeline and context for each job. Operational usage is reflected in the mileage variable, while roadcall_flag indicates whether a breakdown occurred. Labor and notes associated with each service entry are stored in applied_labor and job_note, respectively.

4. Engine Electrical System Rating: The geotab_esr dataset contains 6,722,779 rows and 4 columns, capturing large-scale, time-stamped electrical system data for vehicle units. Each record includes a capture_date, identifying when the data was collected, along with unit_num and device_id, which uniquely identify the vehicle and its associated telematics device. The primary variable of interest, electrical_system_rating, measures the performance or health status of the vehicle's electrical system over time.

5. Battery Roadcalls Data: The battery road calls dataset consists of 59,702 rows and 25 columns, documenting detailed information related to battery-related roadside assistance events. Each record is associated with a specific vehicle, identified by unit_num, and includes vehicle specifications such as category, make_code, model_year, engine_make, engine_model, and corresponding EPA compliance information. The dataset also includes operational features like reefer_flag and liftgate_flag, indicating the presence of refrigeration units or liftgates, which may impact battery usage. Roadside events are described through variables such as roadcall_call_dt (date of the call), roadcall_cause, roadcall_complaint, roadcall_correction, and roadcall_callnum, offering a comprehensive view of each incident. Additional fields such as comp_code_numeric, and comp_code_description provide coded and descriptive classifications of the component or issue involved.

For the binary classification model, we use Dataset 1 (telematics hourly data) and Dataset 3 (battery repair data), which are combined and cleaned before model development. For the LASSO linear regression, outlier detection, and text mining tasks, we utilize Dataset 2 (telematics weekly data), Dataset 3 (battery repair data), Dataset 4 (engine electrical system rating), and Dataset 5 (battery road calls data). These datasets are merged, cleaned, transformed, and encoded as needed before performing the analyses and building the models.

2.4 ANALYTICAL FRAMEWORK

Figure 1 shows the entire analyical framework. The analytics framework for this study is designed to address four key projects: binary classification for battery replacement, LASSO regression for time-to-repair prediction, anomaly detection, and text mining on maintenance job notes. Each stage of the process leverages robust data integration, advanced modeling techniques, and actionable insights for business decision-making.

The process begins with a comprehensive data integration and cleaning phase. Multiple datasets—including telematics hourly and weekly data, battery repair records, engine electrical system ratings, and battery road call logs—are combined to create a unified, high-quality dataset. This involves resolving inconsistencies, handling missing values, and encoding relevant features to ensure accuracy and compatibility across all sources. For the binary classification task, the cleaned telematics hourly data and battery replacement records serve as input for a Random Forest classification model. This algorithm constructs an ensemble of decision

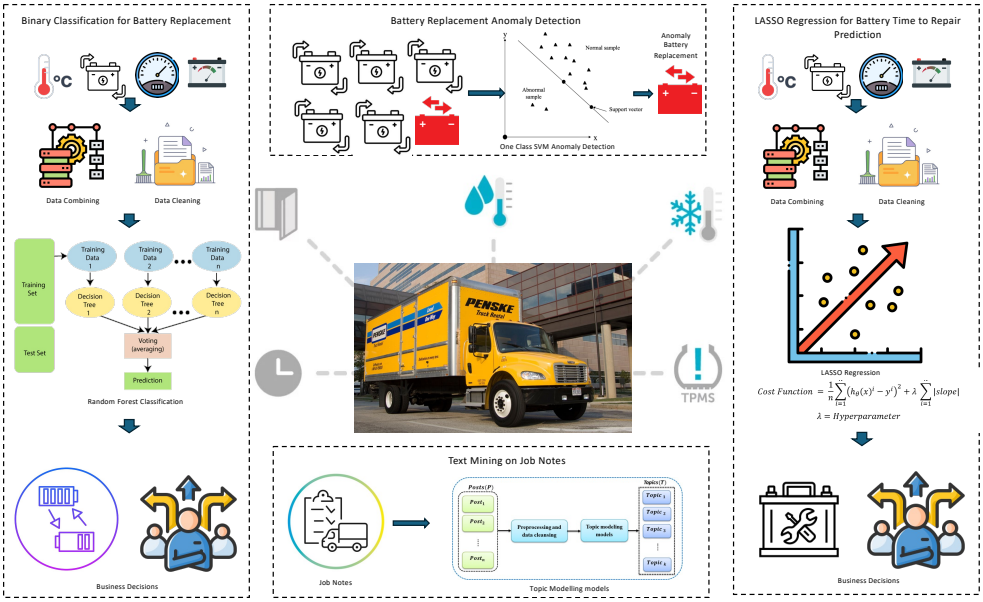


FIGURE 1: ANALYTICAL FRAMEWORK OVERVIEW

trees, each trained on different subsets of the data and features, to predict whether a battery replacement is needed. The individual predictions from these trees are aggregated through majority voting, resulting in a robust and reliable final classification. This approach effectively captures complex, nonlinear relationships within the telematics data, enhances prediction accuracy, and mitigates overfitting. The outputs of this model are directly utilized to identify vehicles at risk of battery failure, supporting proactive maintenance scheduling and reducing operational disruptions. To estimate the expected time until a battery requires repair, LASSO regression is applied using telematics weekly data and battery repair information. LASSO regression is particularly effective in high-dimensional settings, as it performs both variable selection and regularization to improve model interpretability and prevent overfitting. The resulting predictions assist in resource allocation and maintenance planning by forecasting repair timelines more accurately. Anomaly detection is conducted using a One-Class SVM, which identifies unusual or unexpected patterns in battery replacement events. This model learns the normal behavior in the dataset and flags data points that deviate significantly. In parallel, text mining and topic modeling are applied to the free-text job notes associated with each maintenance record. This analysis extracts meaningful patterns and themes from unstructured data, offering additional insights into common repair causes.

3 BINARY CLASSIFICATION MODEL FOR BATTERY REPLACEMENT

3.1 DATA CLEANING AND PREPARATION

Firstly, to establish a comprehensive telematics dataset for advanced analytics, we undertook a systematic process to convert and consolidate raw data originating from multiple sources. The initial step involved aggregating all telematics hourly data files, each stored in Parquet format and distributed across various subfolders. Leveraging a robust and scalable data pipeline developed in Python, every Parquet file was automatically identified and seamlessly integrated into the process. Each file's data was efficiently loaded into memory and appended to a growing collection of data tables. Once all individual files were processed, the entire collection was unified into a single consolidated dataset, representing the full spectrum of telematics records across the designated period. The result is a comprehensive, analysis-ready master dataset, which contains 4,598,840 rows and 30 columns. This consolidated file was then exported in CSV format, making it accessible for subsequent modeling, reporting, and decision-making tasks.

To ensure the reliability and readiness of our analytics pipeline, we implemented a rigorous data cleaning and preprocessing process. The workflow began by merging the telematics dataset with battery replacement records, aligning the data based on unique unit identifiers. To guarantee data quality, we filtered the combined dataset to retain only those records with complete critical information and within the designated analysis window (2 months). Duplicate entries were removed to prevent bias and redundancy in subsequent analyses. Next, we enriched the dataset by engineering several new features, including calculated fields such as duty cycle tiers and miles traveled per day. We also applied logical flagging to indicate battery replacement events, facilitating downstream predictive tasks. Irrelevant columns and those with excessive missing values (greater than 80%) were systematically eliminated to streamline the dataset. For remaining missing data, we adopted a robust imputation strategy: numeric columns were filled using mean values, while categorical variables were completed using the most frequent category. All categorical variables were encoded into machine-readable formats, supporting efficient and accurate model training. As a result of this process, the final cleaned and preprocessed dataset contains 348,794 rows and 39 columns, providing a robust foundation for advanced analytics and machine learning applications.

3.2 EXPLORATORY DATA ANALYSIS

To provide a comprehensive understanding of battery health and maintenance needs, we conducted a thorough data visualization analysis using the cleaned and preprocessed telematics

and maintenance dataset (final dimensions: 348,794 rows × 39 columns). Our visual analytics focused on uncovering key operational patterns and business drivers behind battery replacements. Our initial visualizations featured detailed boxplots of various vehicle operational metrics such as cranking voltage, ambient temperature, odometer readings, and other telematics indicators, categorized by battery replacement outcomes as Figure 2 shows. And Table 1 shows the five-number summaries for the boxplots.

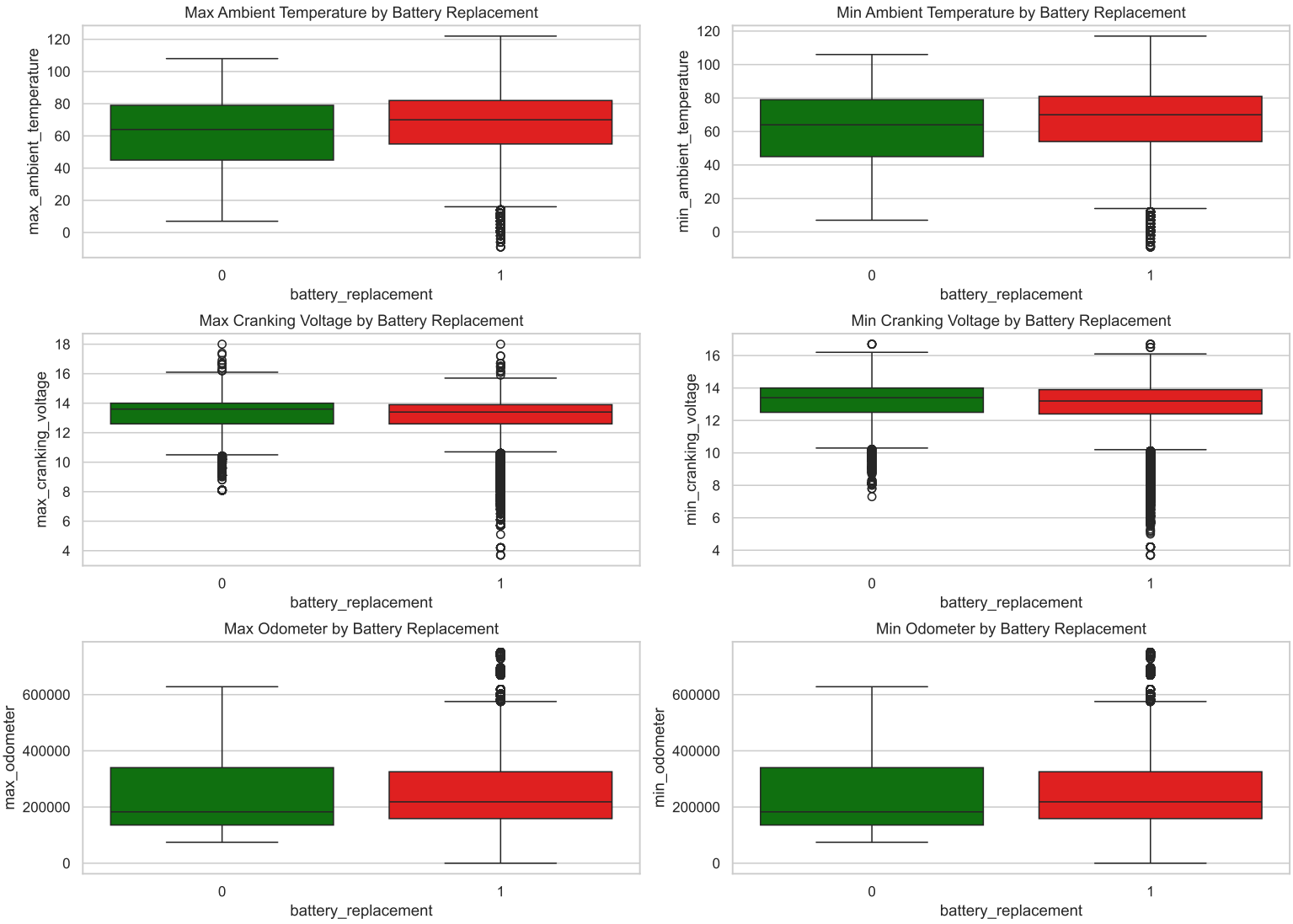


FIGURE 2: BATTERY REPLACEMENT BOXPLOT

Specifically, vehicles needing replacements consistently demonstrate higher odometer readings, with a median maximum odometer reading significantly higher (approximately 218,413 miles) compared to non-replacement vehicles (approximately 182,795 miles). Moreover, replacement vehicles exhibit more extreme values in both ambient temperatures and cranking voltages, indicating operational environments with greater environmental and operational stresses. Particularly, replacement vehicles are subjected to more severe temperature extremes (-9°F to 122°F) and wider voltage variability (minimum cranking voltage as low as 3.7 volts), strongly correlating with battery deterioration.

TABLE 1: FIVE-NUMBER SUMMARIES BY BATTERY REPLACEMENT STATUS

Variable	Replacement	Min	25%	50%	75%	Max
Max Ambient Temperature	Non-Replacement (0)	7.00	45.00	64.00	79.00	108.00
	Replacement (1)	-9.00	55.00	70.00	82.00	122.00
Min Ambient Temperature	Non-Replacement (0)	7.00	45.00	64.00	79.00	106.00
	Replacement (1)	-9.00	54.00	70.00	81.00	117.00
Max Cranking Voltage	Non-Replacement (0)	8.10	12.60	13.60	14.00	18.00
	Replacement (1)	3.70	12.60	13.40	13.90	18.00
Min Cranking Voltage	Non-Replacement (0)	7.30	12.50	13.40	14.00	16.70
	Replacement (1)	3.70	12.40	13.20	13.90	16.70
Max Odometer (k)	Non-Replacement (0)	74.3	135.8	182.8	340.1	628.1
	Replacement (1)	0.0	158.8	218.4	325.5	750.6
Min Odometer (k)	Non-Replacement (0)	74.3	135.8	182.8	340.1	628.1
	Replacement (1)	0.0	158.8	218.4	325.5	750.6

Figure 3 compares average values for several key telematics variables—maximum and minimum odometer readings, miles traveled per day, maximum and minimum ambient temperature, and maximum and minimum cranking voltage—between vehicles that required battery replacement and those that did not. The data shows that vehicles undergoing battery replacement have higher average odometer readings (251,529 for maximum and 251,518 for minimum) compared to non-replacement vehicles (227,351 and 227,344, respectively), indicating that higher cumulative mileage is associated with increased battery failure risk. Similarly, both maximum and minimum ambient temperatures are elevated for replacement vehicles (68.37°F and 67.33°F, respectively) compared to their counterparts (61.99°F and 61.18°F), suggesting that operating in more extreme environments accelerates battery wear.

On the other hand, vehicles that did not require replacement show slightly higher average miles traveled per day (149.41 vs. 141.11), indicating that consistent daily usage does not necessarily increase battery risk, and may even be associated with longer battery life. Additionally, both maximum and minimum cranking voltages are modestly lower in vehicles needing replacement (13.06V and 12.91V) compared to those not replaced (13.27V and 13.15V), underscoring the importance of voltage monitoring for early detection of battery issues.

The actionable insights from the Figure 3 are: (1) Mileage and Environment: Prioritize battery inspections and preventive replacements for vehicles with high odometer readings and those exposed to consistently high or low ambient temperatures. (2) Voltage Monitoring: Implement telematics-based alerts for vehicles trending toward lower cranking voltages to catch and address battery degradation before failure occurs. (3) Utilization Patterns: Consider further analysis into daily usage patterns, as steady daily mileage may correlate with better battery performance.

TABLE 2: FIVE-NUMBER SUMMARY BY BATTERY REPLACEMENT STATUS FOR ENGINE MAKE: CUM

Variable	Replacement	Min	25%	50%	75%	Max
Max Ambient Temperature	Non-Replacement (0)	9.00	54.00	72.00	82.00	108.00
	Replacement (1)	-6.00	55.00	70.00	81.00	122.00
Min Ambient Temperature	Non-Replacement (0)	7.00	52.00	72.00	82.00	106.00
	Replacement (1)	-8.00	54.00	68.00	81.00	117.00
Max Cranking Voltage	Non-Replacement (0)	8.80	12.30	12.90	13.80	16.20
	Replacement (1)	4.20	12.30	13.10	13.90	16.20
Min Cranking Voltage	Non-Replacement (0)	7.30	12.10	12.70	13.80	14.50
	Replacement (1)	4.20	12.00	12.90	13.80	15.20
Max Odometer ($\times 10^3$)	Non-Replacement (0)	74.3	135.7	159.1	214.3	506.5
	Replacement (1)	0.2	151.4	199.9	242.3	728.4
Min Odometer ($\times 10^3$)	Non-Replacement (0)	74.3	135.7	159.0	214.3	506.5
	Replacement (1)	0.1	151.4	199.9	242.3	728.4

TABLE 3: FIVE-NUMBER SUMMARY BY BATTERY REPLACEMENT STATUS FOR ENGINE MAKE: DET

Variable	Replacement	Min	25%	50%	75%	Max
Max Ambient Temperature	Non-Replacement (0)	7.00	39.00	52.00	70.00	104.00
	Replacement (1)	-9.00	55.00	72.00	82.00	118.00
Min Ambient Temperature	Non-Replacement (0)	7.00	37.00	52.00	70.00	99.00
	Replacement (1)	-9.00	54.00	70.00	82.00	113.00
Max Cranking Voltage	Non-Replacement (0)	8.10	13.30	13.90	14.20	18.00
	Replacement (1)	3.70	13.00	13.60	13.90	18.00
Min Cranking Voltage	Non-Replacement (0)	8.10	13.20	13.90	14.20	16.70
	Replacement (1)	3.70	12.80	13.50	13.90	16.70
Max Odometer ($\times 10^3$)	Non-Replacement (0)	85.7	136.9	341.3	367.2	628.1
	Replacement (1)	0.1	186.9	306.5	429.4	750.6
Min Odometer ($\times 10^3$)	Non-Replacement (0)	85.7	136.9	341.3	367.2	628.1
	Replacement (1)	0.1	186.8	306.4	429.4	750.6

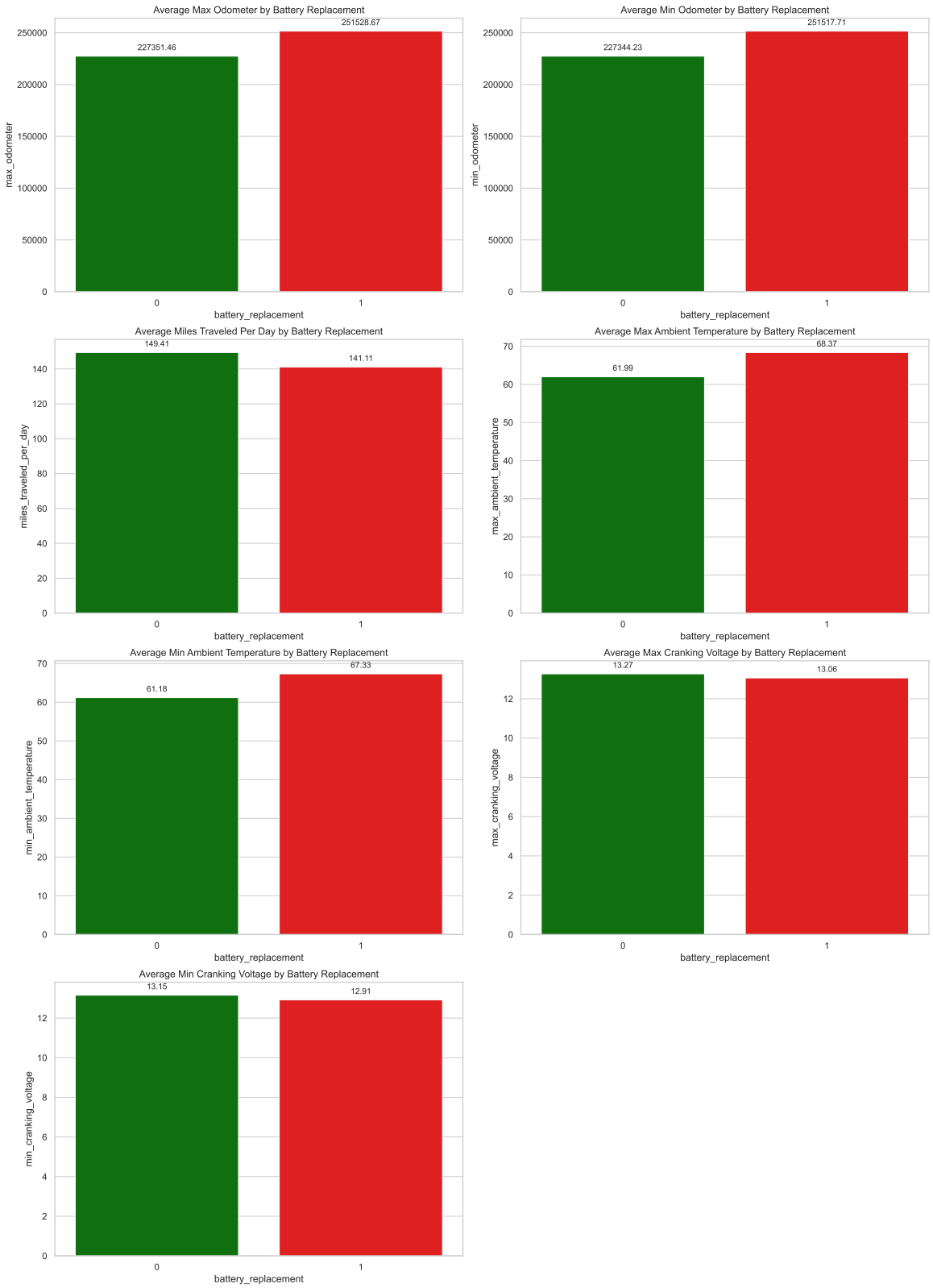


FIGURE 3: AVERAGE VALUES COMPARISON FOR BATTERY REPLACEMENT

The tables 2, 3, and 4 above present detailed five-number summaries for key operational variables by engine make (CUM, DET, and JAS) and battery replacement status. For both the CUM and DET engine makes, vehicles that underwent battery replacement (”1”) consistently exhibit more extreme operational values than those that did not (”0”). Specifically, replace-

TABLE 4: FIVE-NUMBER SUMMARY BY BATTERY REPLACEMENT STATUS FOR ENGINE MAKE: JAS

Variable	Replacement	Min	25%	50%	75%	Max
Max Ambient Temperature	Replacement (1)	12.00	34.00	43.00	52.00	66.00
Min Ambient Temperature	Replacement (1)	12.00	34.00	43.00	52.00	66.00
Max Cranking Voltage	Replacement (1)	9.90	12.90	13.80	14.00	14.00
Min Cranking Voltage	Replacement (1)	8.90	12.90	13.50	14.00	14.00
Max Odometer ($\times 10^3$)	Replacement (1)	225.0	225.6	226.8	227.8	228.5
Min Odometer ($\times 10^3$)	Replacement (1)	225.0	225.6	226.8	227.8	228.5

ment vehicles show: (1) Lower minimum and higher maximum ambient temperatures, suggesting they are exposed to a broader range of environmental stresses. (2) Lower minimum and higher maximum cranking voltages, indicating greater variability and more frequent dips into voltage ranges associated with higher failure risk. (3) Higher maximum and minimum odometer readings, implying that cumulative mileage is a strong predictor of battery failure. In CUM engines, for example, the maximum odometer readings for replacement vehicles reach up to 728k miles compared to 506k for non-replacement vehicles. Similar trends are seen for DET engines. These patterns affirm the role of both usage intensity and operational stress in driving battery replacements. Additionally, replacement groups for both engine makes show broader interquartile ranges (IQR), reflecting more variability in both usage and environmental factors. For the JAS engine make, the dataset only contains vehicles with battery replacements; there is no corresponding non-replacement group within the analyzed two-month time window. The fleet composition would explain why we don't have non failure data points for vehicles with engine make 'JAS'. They contribute to only a 0.003% of the current active fleet at Penske Truck Leasing.

Figure 4 compares the average minimum cranking voltage across three voltage categories—Below 12V, 12V to 14V, and Above 14V—for vehicles that experienced battery replacement and those that did not . The findings reveal that vehicles falling in the "Below 12V" category have the lowest average minimum cranking voltages (11.03V for replacements and 11.29V for non-replacements), signaling greater risk for battery failure in this group. In contrast, vehicles in the higher voltage categories ("12V to 14V" and "Above 14V") exhibit noticeably higher average minimum voltages, with only marginal differences between the replacement and non-replacement groups. The key findings are: (1) Cranking voltage below 12V is strongly associated with an increased likelihood of battery replacement. The average minimum cranking voltage is lowest for vehicles needing replacement, suggesting early warning potential for batteries in this voltage range. (2) In the higher voltage categories, the average minimum voltage is similar for both groups, indicating a reduced risk of imminent battery failure when voltages

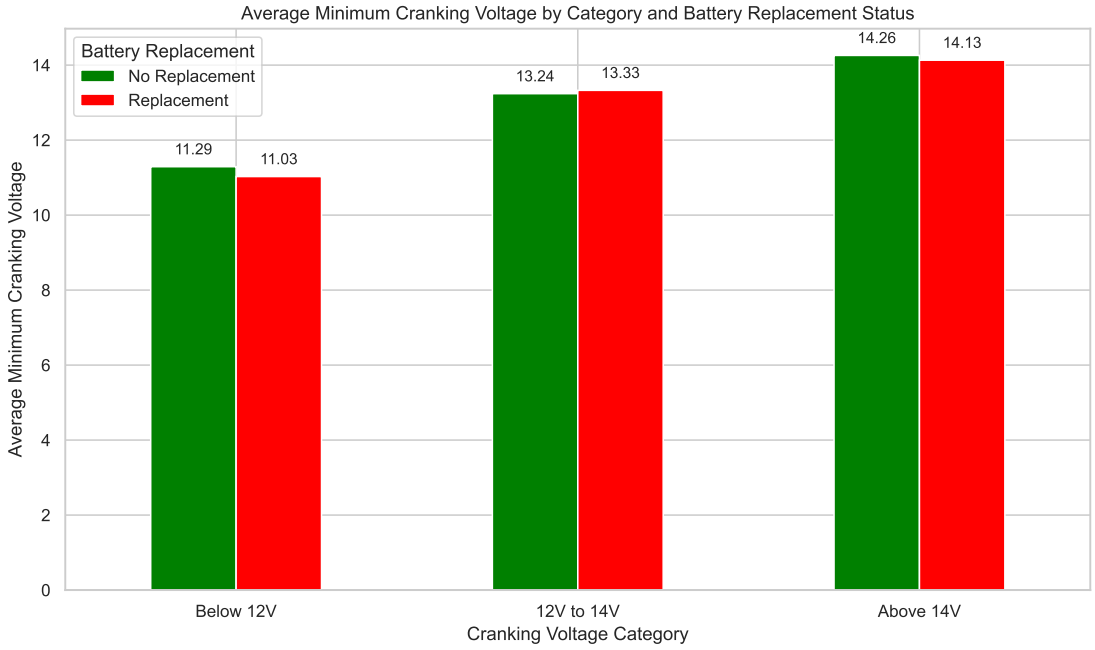
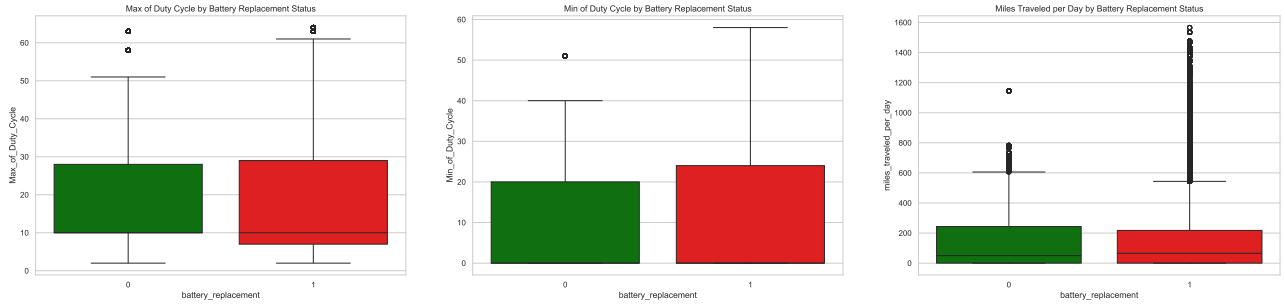


FIGURE 4: CRANKING VOLTAGE CATEGORY VS. BATTERY REPLACEMENT

are consistently above 12V.



(a) MAXIMUM OF DUTY CYCLE **(b)** MINIMUM OF DUTY CYCLE **(c)** MILES TRAVELED PER DAY

FIGURE 5: DUTY CYCLE AND DAILY MILES BY BATTERY REPLACEMENT STATUS

Figures 5 are the five-number summaries compare operational metrics for vehicles with ("1") and without ("0") battery replacements. For the maximum duty cycle, both groups have similar ranges (min: 2, max: 63–64), but the upper quartile (Q3) and median are slightly higher for the replacement group, suggesting that vehicles needing replacement are more likely to experience higher peaks in maximum duty cycle. For the minimum duty cycle, both groups have a median and Q1 of zero, but the upper quartile and maximum are higher in the replacement group (Q3: 24.01 vs. 20.01; max: 58.01 vs. 51.01), indicating that even the least-used periods are somewhat more demanding for vehicles that eventually require battery replacement. If the higher cranking voltages in the 3rd quartile / median for units with failures within a duty cycle, those could be attributed to cranking voltages captured during or after jump start procedures. When examining miles traveled per day, vehicles requiring battery replacement have a higher

upper range (Q3: 217.7 vs. 243.3; max: 1566.1 vs. 1145.4) and a higher median (65.6 vs. 50.0) compared to non-replacement vehicles. This suggests that both high-intensity usage (greater maximums) and sustained regular use (higher medians and Q3s) are associated with increased risk for battery failure.

It shows that vehicles exposed to more intense or sustained duty cycles, and those with higher daily mileage, are at increased risk for battery issues.

3.3 BINARY CLASSIFICATION MODELING AND EVALUATION

The primary objective of this analysis is to predict whether a vehicle will require battery replacement, leveraging the rich set of telematics hourly data as predictor variables (X) and the battery replacement status as the response variable (y). The predictor variables include a wide range of operational and environmental features continuously recorded from the fleet, such as cranking voltage, ambient temperature, odometer readings, duty cycle, and miles traveled per day. The response variable y is a binary indicator: $y = 1$ if the vehicle experienced a battery replacement event, and $y = 0$ otherwise.

To address this binary classification task, we trained and compared multiple machine learning models, including Random Forest, Decision Tree, and Gradient Boosting classifiers. The modeling workflow comprised the following steps:

- **Data Preparation:** The consolidated and cleaned dataset was split into training and validation subsets, ensuring that the target variable distribution was balanced across splits.
- **Feature Engineering:** All relevant telematics features were standardized or encoded as needed to optimize model performance and interpretability.
- **Model Training:** Each candidate model was trained on the training set using the telematics features as input (X) and battery replacement status (y) as the target.
- **Model Validation:** Model predictions were evaluated on the validation set, and key classification metrics were computed to compare performance.
- **Model Selection:** The best-performing model was selected based on a combination of metrics, prioritizing balanced accuracy, precision, recall, F1-score, and ROC-AUC.

Formally, let $X \in R^{n \times p}$ denote the matrix of n samples and p telematics features, and $y \in \{0, 1\}^n$ the corresponding vector of battery replacement outcomes. The classification models learn a function $f : X \to y$ that predicts the likelihood of battery replacement events.

The evaluation of each model was conducted using the following metrics, here, TP (True Positives) represents the number of vehicles correctly predicted as requiring battery replacement, TN (True Negatives) denotes vehicles correctly predicted as not requiring replacement, FP (False Positives) refers to vehicles incorrectly predicted as requiring replacement when

they do not, and FN (False Negatives) indicates vehicles incorrectly predicted as not requiring replacement when they actually do. :

- **Accuracy:**
$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$
- **Precision:**
$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$
- **Recall:**
$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$
- **F1-score:**
$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$
- **AUC (Area Under the ROC Curve):** is a performance metric used in binary classification problems. It summarizes a model's ability to distinguish between positive and negative instances across all possible classification thresholds.

All the evaluation metrics values are between 0 and 1, where 0 indicates worst performance and 1 indicates perfect performance. The table 5 summarizes the performance of each model on the validation dataset: Based on the validation results, the Random Forest classifier

TABLE 5: CLASSIFICATION MODEL PERFORMANCE

Model	Accuracy	Precision	Recall	F1-score	AUC
Random Forest	0.9908	0.9953	0.9946	0.9950	0.9995
Decision Tree	0.9908	0.9962	0.9937	0.9949	0.9894
Gradient Boosting	0.9723	0.9720	0.9985	0.9851	0.9866

was selected as the best model. Its outstanding accuracy, F1-score, and ROC-AUC demonstrate that the model can reliably predict battery replacement risk using telematics hourly data. Therefore, we selected the Random Forest model for future use and saved the trained model so that Penske can readily deploy it as needed.

To further validate the robustness of the selected Random Forest model, we evaluated its performance on a randomly selected hold-out test dataset. The results demonstrate consistently high predictive accuracy and reliability, as summarized in Table 6.

These results confirm the model's strong generalization capability and its suitability for reliable deployment in Penske's operational environment.

TABLE 6: RANDOM FOREST MODEL PERFORMANCE ON TEST SET

Accuracy	Precision	Recall	F1-score	AUC
0.9913	0.9952	0.9953	0.9953	0.9995

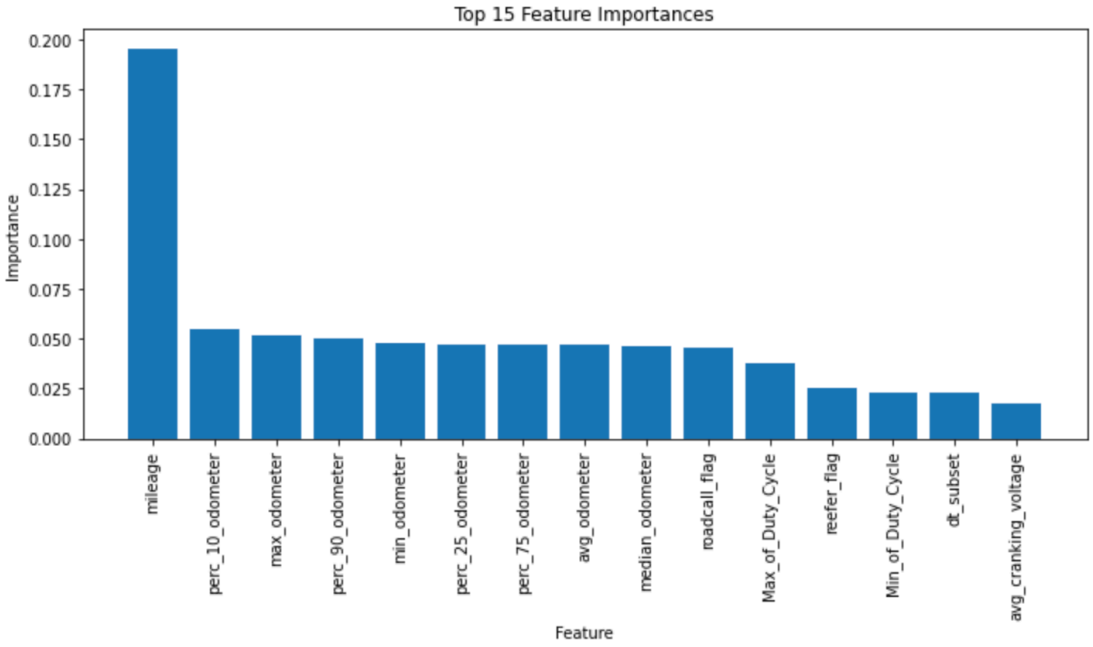


FIGURE 6: TOP 15 FEATURE RANKINGS FOR BATTERY REPLACEMENT CLASSIFICATION

3.4 MODEL INTERPRETATION AND BUSINESS IMPLICATIONS

Figure 6 illustrates the top 15 most influential features identified by the Random Forest model in predicting battery replacement events. The most dominant predictor is mileage, indicating that the total distance a vehicle has traveled plays the most significant role in assessing battery health and failure risk. Other key contributors include several odometer-based percentile metrics (such as the 10th, 25th, 75th, and 90th percentiles of odometer readings), the maximum and minimum odometer readings, and the average odometer. These collectively highlight that various aspects of cumulative distance and usage patterns provide strong predictive signals for battery replacement needs. Additional features with notable importance are duty cycle statistics (maximum, minimum, and median), which reflect the operational intensity and variability in engine usage, as well as roadcall_flag. Environmental or operational flags such as reefer and the categorization using miles travelled for every hour of engine runtime also contribute meaningfully. Finally, average cranking voltage—a direct measure of battery performance—is also identified as a relevant risk indicator.

To interpret the predictions made by our Random Forest model, we employed LIME (Local Interpretable Model-agnostic Explanations). LIME is a powerful tool that helps explain the output of any black-box classifier by approximating it locally with a simpler, interpretable model.

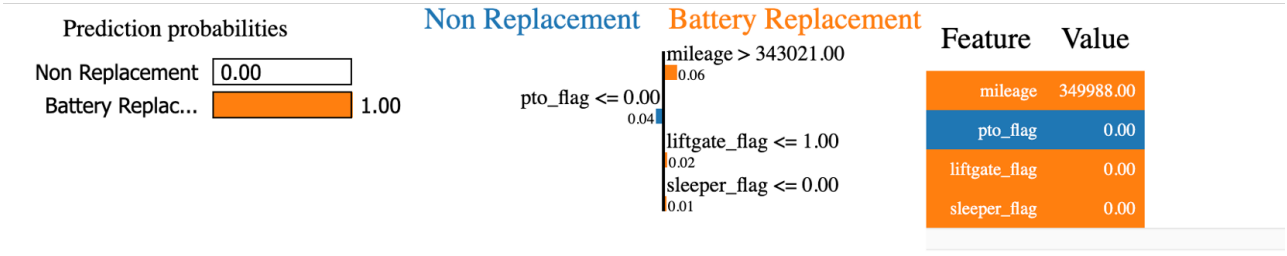


FIGURE 7: MODEL INTERPRETATION USING LIME

As Figure 7 shows, the model assigned a probability of 1.00 to the "Battery Replacement" class, primarily due to the vehicle's high mileage (exceeding 343,021 miles), other features such as `pto_flag`, `liftgate_flag`, and `sleeper_flag` were also have influential.

The Random Forest classification model demonstrated outstanding predictive performance, accurately identifying vehicles likely to require battery replacement. Key operational indicators such as high mileage and specific telematics metrics emerged as significant predictors, enabling Penske to proactively target vehicles most at risk of battery failures. Leveraging LIME for interpretability further enhances the practical value of this predictive tool by clearly illustrating how individual decisions are made, increasing stakeholder confidence in adopting machine-learning-driven strategies. Penske is therefore advised to integrate this model into their regular maintenance and fleet management practices, enabling precise, proactive maintenance scheduling and substantial reductions in vehicle downtime and operational disruptions. The deployment of this predictive solution represents a meaningful step forward in optimizing resource allocation, improving fleet reliability, and ultimately enhancing Penske's service delivery and customer satisfaction. Here are some recommendations:

- **Real-Time Telematics Alerts:** Implement a real-time monitoring system linked to the model, triggering automated alerts when vehicles cross key thresholds, such as excessive mileage or low cranking voltage, thus enabling timely maintenance interventions.
- **Operational Policy Adjustments:** Leverage insights from the model and LIME interpretations to inform operational policies. For example, consider mileage-based thresholds or environmental conditions in fleet rotation policies to extend battery life and optimize vehicle utilization.
- **Inventory Optimization:** Align battery inventory and procurement strategies with model predictions, ensuring availability of replacement batteries where and when they are most likely needed, thus minimizing inventory carrying costs and reducing delays.
- **Continuous Model Refinement:** Regularly retrain and update the predictive model with fresh telematics data to capture emerging patterns and ensure sustained predictive accuracy, keeping Penske at the forefront of fleet management innovation.

4 TIME-TO-REPAIR PREDICTION USING LASSO REGRESSION

4.1 DATA CLEANING AND PREPARATION

We began our analysis by preparing and cleaning the datasets. First, we removed outliers and handled missing data. We then merged the battery repair and engine datasets using a left join, which allowed us to retain all battery repair records—even those without corresponding engine data—since our primary focus was on battery-related issues. This initial merge resulted in a dataset containing 2,277,122 rows and 25 columns. Upon further inspection, we found duplicate records, so we removed them as the final step before analysis. This left us with 1,616,962 rows and 25 columns.

To ensure the dataset was suitable for linear regression, we checked the percentage of missing data to prevent bias or model errors. During this process, we identified that the "unit engine epa type" column, which could have been a useful predictor, had more than 99% missing values and was therefore excluded from further analysis. In contrast, the "dt_subset" column, representing categorization using miles travelled for every hour of engine runtime, had 16% missing values. Rather than dropping this column, we imputed the missing values with a constant ("C") to preserve the information. We also created a new variable, "time_to_repair," by subtracting the incident date from the job start date. This variable served as the target for our regression analysis, allowing us to estimate how long before a truck experience an unplanned repair.

To enhance our regression model, we engineered a new feature called "duty cycle," which reflects how intensively a truck is used based on its engine model, tier ("dt_tier"), and subset ("dt_subset"). Using a predefined dictionary, we matched each combination to a specific range and created two new columns: "max_of_duty_cycle" and "min_of_duty_cycle." These additions provided deeper insights into how engine usage might affect battery issues. We also streamlined the dataset by removing columns with only one unique value, using a function we developed to identify and drop such columns. Additionally, we converted non-numeric data types as needed to ensure compatibility with linear regression.

4.2 EXPLORATORY DATA ANALYSIS

Once the data preparation was complete, we generated a correlation matrix to examine the relationships between variables. By visualizing this matrix as a heatmap, we were able to quickly identify which variables were strongly or weakly related. In the heatmap, darker or more intense colors represent stronger correlations, whether positive or negative. This step was essential for selecting useful predictors for our linear regression model and for identifying variables that might be redundant or unrelated to the target variable.

The correlation matrix 8 offers a detailed summary of the linear relationships among the

variables in our dataset. Analysis of this matrix revealed several important patterns: First, there are extremely high positive correlations among various odometer-related variables. For example, "median_odometer," "avg_odometer," "perc_90_odometer," and related metrics all show correlation coefficients very close to 1. This indicates these odometer-based features are essentially measuring the same underlying attribute, leading to a high degree of redundancy in the data. Other findings include moderate correlations among certain feature flags. For example, "dt_tier" shows negative correlations with several odometer variables (approximately -0.44), as well as with "sleeper_flag" and "mileage." This suggests that truck classification and configuration (such as the presence of a sleeper cab) are inversely related to how far the vehicles have traveled and certain usage patterns. Such relationships highlight the interconnected nature of truck features and usage profiles, underscoring the importance of carefully selecting predictors for analysis.

When focusing on our main outcome, "time_to_repair," we observe that it has only weak correlations with all other variables (most near zero). This indicates that the predictors in our dataset do not have a strong linear relationship with repair time, making it challenging to model and predict repair duration based solely on these features.

A critical issue revealed by the correlation matrix is the presence of multicollinearity. The odometer variables are so highly correlated with one another that including all of them as predictors could lead to unstable regression coefficients and difficulty in interpreting the true influence of each variable. Multicollinearity can inflate standard errors, reduce statistical power, and ultimately result in misleading model interpretations. To address multicollinearity and redundancy, we employed the LASSO regression technique. LASSO regularizes the regression by penalizing large coefficients, effectively shrinking less relevant variables toward zero and automatically selecting a more informative subset of predictors. This process improves model interpretability and robustness, allowing us to focus on the most meaningful features while mitigating the risk of overfitting and misinterpretation.

Table 7 summarizes the top positive and negative correlations among variables in the dataset. The strongest positive correlations are observed between various odometer-related features, such as "median_odometer" and "avg_odometer," as well as between different percentile measures of odometer readings. These high correlation coefficients, all above 0.99, indicate that these variables are essentially capturing the same information and may be redundant if included together in predictive models. On the negative side, "dt_tier" exhibits moderate negative correlations (around -0.44) with several odometer metrics, suggesting that truck classification tiers are inversely related to cumulative distance traveled. Additionally, "sleeper_flag" and "dt_tier" show a negative relationship, while "liftgate_flag" and "sleeper_flag" are also negatively correlated. These patterns highlight both potential redundancy among variables and important associations between truck features, emphasizing the need for careful feature selection and multicollinearity management in subsequent analyses.

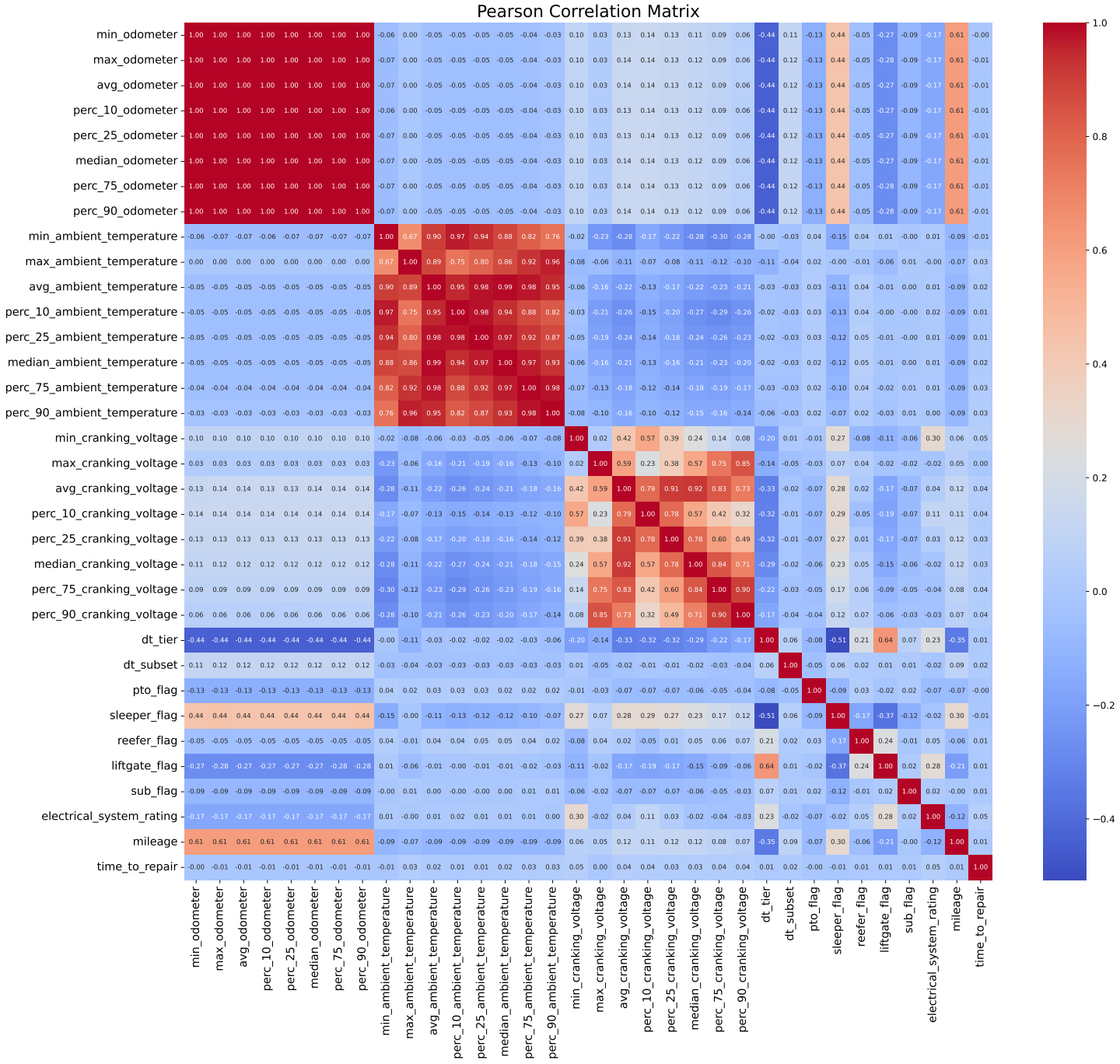


FIGURE 8: PEARSON CORRELATION MATRIX BETWEEN DIFFERENT VARIABLES

4.3 LASSO REGRESSION MODELING AND EVALUATION

LASSO (Least Absolute Shrinkage and Selection Operator) regression is a regularization technique that enhances the prediction accuracy and interpretability of statistical models by imposing a penalty on the absolute value of the regression coefficients. This penalty encourages sparsity, effectively performing variable selection by shrinking some coefficients to exactly zero. LASSO regression is particularly advantageous in this context because it performs automatic feature selection and helps to mitigate multicollinearity among the predictor variables.

In this study, the predictor variables **X** consist of both engine system rating data and telem-

TABLE 7: TOP POSITIVE AND NEGATIVE CORRELATIONS AMONG VARIABLES

Type	Variable 1	Variable 2	Correlation
Positive	median_odometer	avg_odometer	0.999993
Positive	perc_90_odometer	perc_75_odometer	0.999985
Positive	perc_25_odometer	perc_10_odometer	0.999963
Positive	perc_75_odometer	median_odometer	0.999825
Positive	perc_90_odometer	median_odometer	0.999809
Positive	perc_75_odometer	avg_odometer	0.999786
Positive	perc_25_odometer	avg_odometer	0.999785
Positive	perc_90_odometer	avg_odometer	0.999774
Positive	perc_90_odometer	max_odometer	0.999764
Positive	perc_10_odometer	avg_odometer	0.999748
Negative	sleeper_flag	dt_tier	-0.507467
Negative	dt_tier	perc_90_odometer	-0.440535
Negative	dt_tier	max_odometer	-0.440411
Negative	dt_tier	perc_75_odometer	-0.440206
Negative	dt_tier	median_odometer	-0.439602
Negative	dt_tier	avg_odometer	-0.439582
Negative	dt_tier	perc_25_odometer	-0.438759
Negative	dt_tier	perc_10_odometer	-0.438458
Negative	dt_tier	min_odometer	-0.436650
Negative	liftgate_flag	sleeper_flag	-0.370975

atics data collected. The response variable y is defined as *time_to_repair*, which represents the duration (in days) between the vehicle's actual repair date and the date on which the corresponding engine system rating and telematics data were reported. The LASSO regression estimates the coefficient vector $\boldsymbol{\beta}$ by solving the following optimization problem:

$$\hat{\boldsymbol{\beta}}^{\text{LASSO}} = \arg\min_{\boldsymbol{\beta}} \left\{ \frac{1}{2n} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\} \tag{5}$$

where:

- n is the number of observations.
- p is the number of predictor variables (features from engine system ratings and telematics).
- y_i is the time_to_repair for the i-th vehicle.
- x_{ij} is the value of the j-th predictor for the i-th vehicle.

- β_0 is the intercept term.
- $\boldsymbol{\beta} = (\beta_1, \beta_2, \ldots, \beta_p)$ are the regression coefficients.
- $\lambda \geq 0$ is a regularization parameter that controls the strength of the penalty.

The regularization term $\lambda \sum_{j=1}^{p} |\beta_j|$ encourages sparsity in the coefficient estimates, allowing the model to select only the most relevant features from the engine system rating and telematics data for predicting the time required for vehicle repair.

To evaluate the predictive performance of the LASSO regression model, we use the Root Mean Squared Error (RMSE) metric. The RMSE is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{6}$$

where:

- n is the total number of observations in the test set,
- y_i is the actual time_to_repair for the i-th vehicle,
- $\hat{y}_i$ is the predicted time_to_repair for the i-th vehicle.

In this specific case, RMSE represents the average magnitude of the prediction errors between the actual and predicted time required to repair vehicles, measured in days. A lower RMSE value indicates that the model provides more accurate predictions of the repair time based on the engine system rating and telematics data. In this study, the average RMSE obtained is 17 days, which means that, on average, the predicted time_to_repair differs from the actual time by approximately 17 days for each vehicle. This provides a measure of the typical prediction error when using the engine system rating and telematics data to how long before a truck experience an unplanned repair.

4.4 MODEL INTERPRETATION AND BUSINESS IMPLICATIONS

Table 8 presents the top ten positive and negative features influencing the predicted time to repair, where the outcome variable is defined as the number of days between the telematics data/engine system rating reading and the commencement of the actual repair job. Features with positive coefficients, such as `min_odometer`, `perc_10_ambient_temperature`, and `perc_90_cranking_voltage`, are associated with a longer wait time before repairs are initiated. This may indicate that vehicles with higher minimum odometer readings or higher low-end ambient temperatures are more likely to experience delayed repair actions, possibly due to less apparent or urgent issues, or more challenging fault detection. In contrast, features with negative coefficients, such as `max_odometer`, `min_ambient_temperature`, and `max_cranking_voltage`,

TABLE 8: TOP 10 POSITIVE AND NEGATIVE LASSO COEFFICIENTS

Top Positive Feature	Coefficient	Top Negative Feature	Coefficient
min_odometer	4.68	max_odometer	-5.16
perc_10_ambient_temperature	2.56	min_ambient_temperature	-3.14
perc_90_cranking_voltage	2.12	max_cranking_voltage	-1.86
perc_90_ambient_temperature	1.93	perc_25_ambient_temperature	-1.17
median_ambient_temperature	1.28	max_ambient_temperature	-0.89
min_cranking_voltage	0.85	sleeper_flag	-0.39
dt_subset	0.61	perc_10_cranking_voltage	-0.30
mileage	0.42	perc_75_cranking_voltage	-0.20
electrical_system_rating	0.38	liftgate_flag	-0.14
median_cranking_voltage	0.36	sub_flag	-0.08

are linked to shorter delays between the recording of telematics data and the start of repairs. This suggests that vehicles with these characteristics may trigger faster interventions, perhaps because these metrics reflect more severe, easily detected, or operationally critical conditions.

Categorical variables such as `sleeper_flag`, `liftgate_flag`, and `sub_flag` also have negative coefficients, indicating that vehicles with these configurations tend to move more quickly from the point of telematics alert to repair action, possibly due to established maintenance routines or prioritization policies. These insights enable fleet managers and maintenance teams to better understand the factors that influence repair scheduling and response times. By targeting features associated with longer repair initiation intervals, businesses can refine their predictive maintenance strategies and reduce unnecessary downtime, ultimately improving operational efficiency and service reliability.

5 OUTLIER DETECTION IN BATTERY REPAIR DATA

The primary objective of this project is to detect anomalous battery replacement events within a fleet, utilizing high-dimensional telematics data. Identification of atypical cases enables proactive investigation, root-cause analysis, and targeted operational improvements. This approach supports predictive maintenance, cost control, and enhanced fleet reliability.

5.1 OUTLIER DETECTION METHODOLOGY

We consolidated all available battery replacement records along with corresponding telematics data for each unit. All numerical features were standardized using z-score normalization to ensure scale-invariant modeling. For each feature x, the standardized value z is computed as:

$$z = \frac{x - \mu}{\sigma} \tag{7}$$

where μ and σ are the sample mean and standard deviation of the feature, respectively.

Unlike traditional supervised learning, anomaly detection often faces a scarcity or total absence of labeled anomalous cases. One-Class SVM is an unsupervised method designed to

learn the support of the data distribution, effectively distinguishing "normal" events from outliers. In a mathematical way, given a set of standardized feature vectors $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$, the One-Class SVM algorithm aims to find a decision function f that captures the region where the majority of data reside. The primal optimization problem is formulated as:

$$\min_{\mathbf{w},\rho,\xi_i} \quad \frac{1}{2}\|\mathbf{w}\|^2 + \frac{1}{\nu n}\sum_{i=1}^{n}\xi_i - \rho \tag{8}$$

$$\text{subject to} \quad (\mathbf{w}\cdot\phi(\mathbf{x}_i)) \geq \rho - \xi_i, \quad \xi_i \geq 0, \quad i = 1, \ldots, n \tag{9}$$

where:

- $\phi(\cdot)$ is a mapping to a high-dimensional feature space (induced by the kernel),
- $\nu \in (0, 1]$ sets an upper bound on the fraction of anomalies and a lower bound on the fraction of support vectors,
- ξ_i are slack variables permitting some data to be classified as outliers,
- ρ is the offset parameter.

The corresponding decision function for a new observation $\mathbf{x}$ is:

$$f(\mathbf{x}) = \text{sign}\left((\mathbf{w}\cdot\phi(\mathbf{x})) - \rho\right) \tag{10}$$

Observations for which $f(\mathbf{x}) < 0$ are considered anomalies. The model is trained using the Radial Basis Function (RBF) kernel:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2\right) \tag{11}$$

where γ is a hyperparameter controlling the kernel width.

After fitting the One-Class SVM on the standardized feature matrix, each battery replacement record is classified as either a typical event ($f(\mathbf{x}) \geq 0$) or an anomaly ($f(\mathbf{x}) < 0$). Anomalous events are then flagged for further review, with details such as unit number and service date extracted for operational follow-up.

5.2 VISUALIZING ANOMALIES

Using the One-Class SVM, we are able to identify some anomalies battery replacement incidents. Figure 9 visualizes the results of the anomaly detection process applied to battery replacement data. The plot displays each battery replacement observation projected onto the first two principal components (PC1 and PC2), obtained through Principal Component Analysis (PCA). Observations are colored according to the outlier labels assigned by the One-Class SVM algorithm: yellow points represent typical (inlier) events, while purple points indicate events classified as anomalies (outliers). The visualization demonstrates that outlier events are

dispersed around the periphery of the data cloud, confirming that the One-Class SVM effectively identifies observations that deviate from the dominant patterns in the telematics dataset. This approach enables efficient separation of anomalous maintenance events, facilitating targeted investigation and operational intervention.

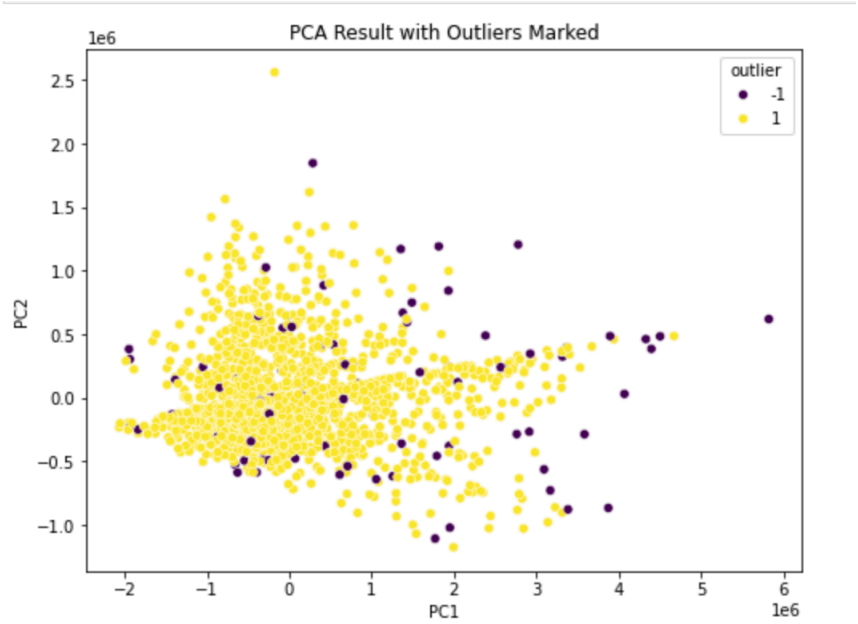


FIGURE 9: VISUALIZATION OF BATTERY REPLACEMENT ANOMALIES VIA PCA

Figure 10 illustrates the ten most influential features for identifying outlier battery replacement events, as determined by a Random Forest classifier. The features, listed in descending order of importance, predominantly represent vehicle telematics data recorded several weeks prior to the battery replacement. For example, perc_75_cranking_voltage_8weekago (feature importance: 0.0215) refers to the 75th percentile of cranking voltage measured eight weeks before the replacement event, which is the single most important predictor. Similarly, features such as perc_10_odometer_5weekago, min_cranking_voltage_7weekago, and max odometer 5weekago indicate odometer and cranking voltage statistics captured five to seven weeks before replacement.

Other significant features include various percentiles and medians of cranking voltage, odometer readings, and ambient temperature, all measured four to eight weeks before the event. The prevalence of these time-lagged features highlights that early deviations in vehicle performance—weeks in advance—are strongly associated with anomalous battery replacements. This temporal insight supports proactive monitoring: by focusing on key trends in the weeks leading up to a battery replacement, it is possible to flag units at higher risk of atypical failure and intervene earlier to prevent costly downtime.

Table 9 provides a detailed comparison of mean values for the ten most influential telematics features used to detect anomalous battery replacements. Consistently, vehicles identified as outliers exhibited notably different patterns compared to typical (non-outlier) cases.

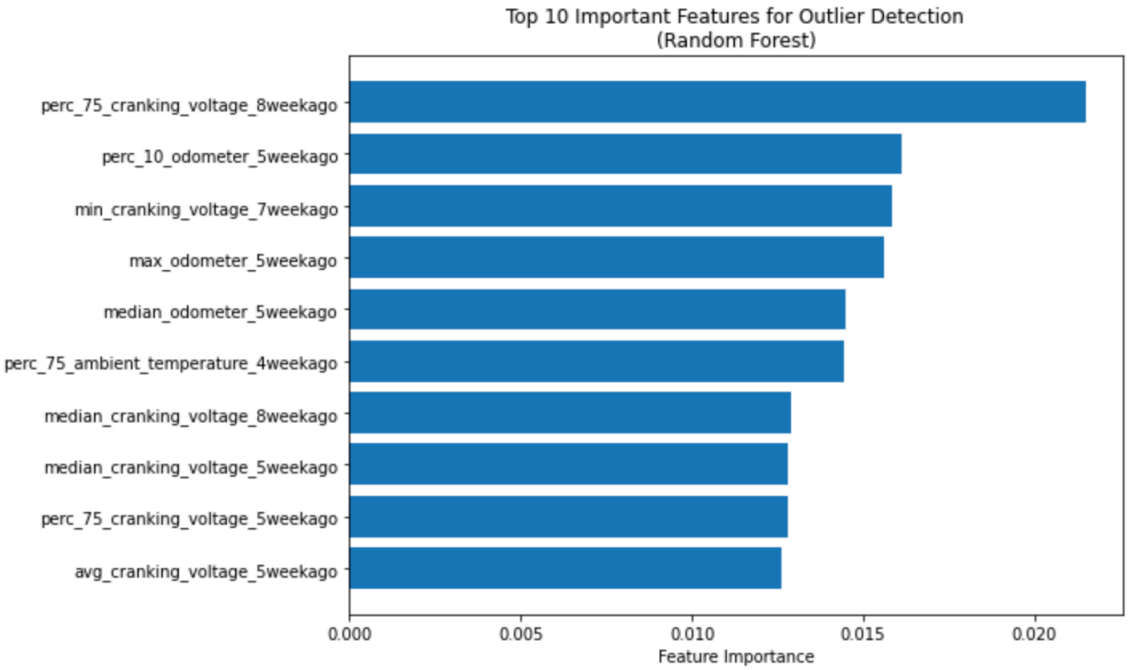


FIGURE 10: TOP 10 IMPORTANT FEATURES FOR OUTLIER DETECTION

Specifically, outlier events showed significantly higher odometer readings five weeks prior to replacement—for example, the average perc_10_odometer_5weekago value for outliers (380,084.51 miles) far exceeded that of non-outliers (252,365.24 miles), indicating higher usage intensity associated with anomalous replacements. Similarly, voltage-related features like perc_75_cranking _voltage_8weekago had lower average values among outliers (12.81 volts) compared to non-outliers (13.42 volts), suggesting reduced battery performance weeks before the replacement event.

The boxplot distributions in Figure 11 further reinforce these observations by illustrating the distinct separation and variability of these key features between the groups. For instance, cranking voltage metrics clearly demonstrate lower median values and increased variability for outliers, reflecting deteriorating electrical performance. Conversely, odometer-based metrics exhibit higher median values and a broader spread among outliers, signifying greater mileage accumulation preceding atypical battery replacements.

5.3 IMPACT ASSESSMENT AND REMEDIATION INSIGHTS

The analysis provides compelling evidence of distinct telematics patterns that precede anomalous battery replacement events, significantly impacting fleet maintenance efficiency and operational costs. The identification of odometer readings and cranking voltage metrics as key indicators—specifically, elevated odometer usage (average odometer exceeding 380,000 miles among outliers) and declining cranking voltage values weeks before replacement. Here are some remediation insights and recommendations:

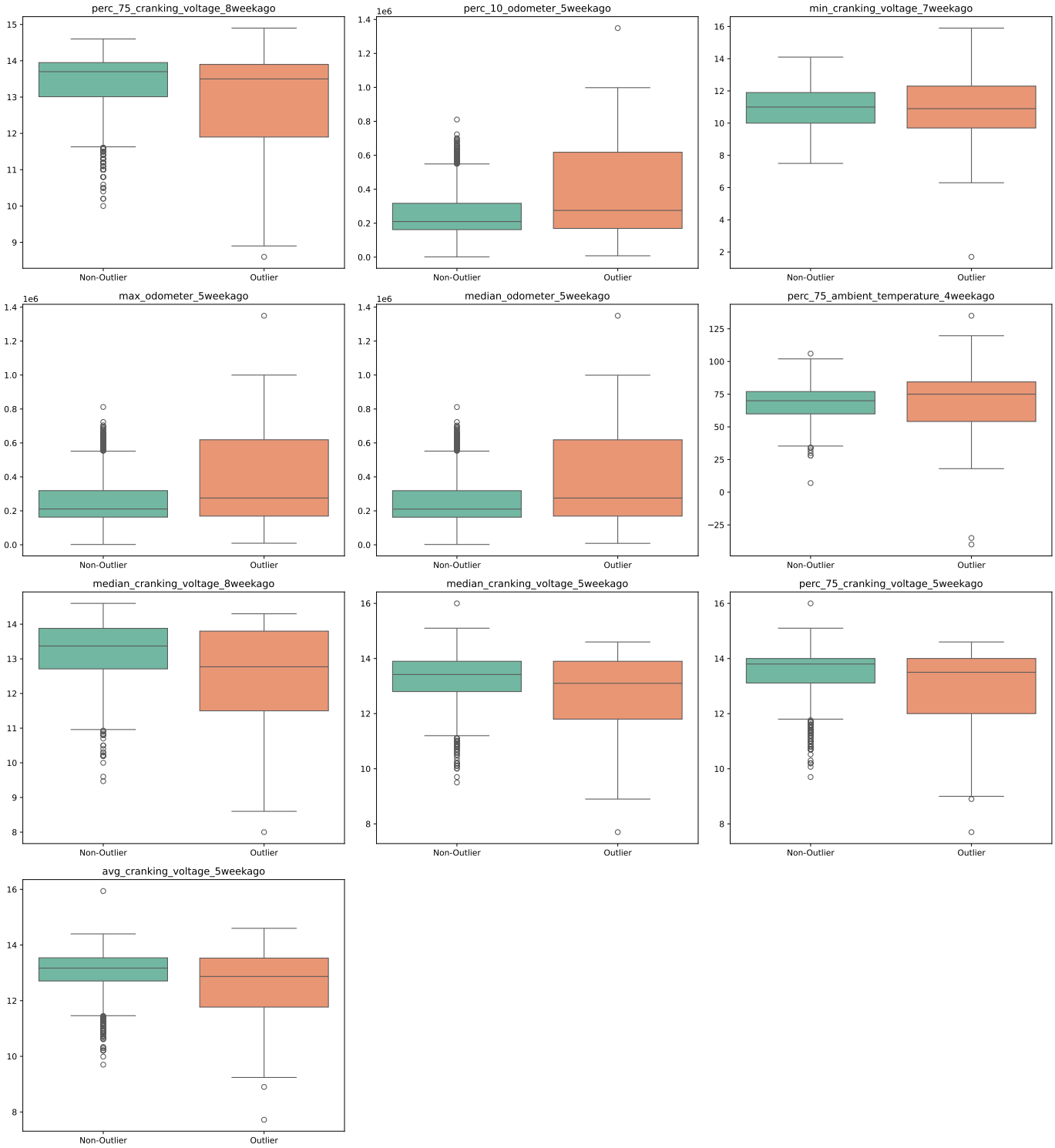


FIGURE 11: TOP 10 FEATURE DISTRIBUTIONS BY OUTLIER STATUS

- Enhanced Proactive Maintenance Scheduling: Penske can integrate the predictive signals from high-impact telematics features into its maintenance scheduling system. Specifically, units exhibiting increased odometer readings or decreased cranking voltage values several weeks prior to their scheduled maintenance should receive prioritized inspections or preventive maintenance interventions to mitigate the risk of unexpected fail-

TABLE 9: MEAN VALUES OF TOP 10 FEATURES BY OUTLIER STATUS

Feature	Mean (Non-Outlier)	Mean (Outlier)
perc_75_cranking_voltage_8weekago	13.42	12.81
perc_10_odometer_5weekago	252,365.24	380,084.51
min_cranking_voltage_7weekago	11.02	10.84
max_odometer_5weekago	253,605.05	380,585.34
median_odometer_5weekago	253,233.27	380,316.63
perc_75_ambient_temperature_4weekago	68.36	69.65
median_cranking_voltage_8weekago	13.17	12.46
median_cranking_voltage_5weekago	13.20	12.63
perc_75_cranking_voltage_5weekago	13.45	12.80
avg_cranking_voltage_5weekago	13.03	12.49

ures.

- Operational Policy Adjustment: Penske can adjust vehicle deployment and usage guidelines, particularly for high-mileage vehicles identified as being at higher risk. Vehicles flagged by these predictive metrics should potentially undergo accelerated maintenance cycles or be rotated strategically to manage workload distribution and avoid costly emergency battery replacements.

6 TEXT MINING ON JOB NOTES

6.1 DATA PREPROCESSING AND TOKENIZATION

The job notes data comprises unstructured text, necessitating comprehensive preprocessing to ensure analytical robustness. We implemented the following preprocessing pipeline:

1. **Normalization:** Conversion of all textual characters to lowercase to ensure uniformity.
2. **Punctuation and Numeric Removal:** Removal of numeric digits, punctuation marks, and special symbols to retain purely textual information.
3. **Tokenization:** Segmentation of textual strings into discrete tokens. Mathematically, for a document d:
$$d \rightarrow \{w_1, w_2, w_3, \ldots, w_n\}$$
where each token w_i represents a meaningful word or symbol in the context.
4. **Stop-word Removal:** Eliminating frequently occurring but contextually irrelevant words using standard English stop-word lexicons.

5. **Lemmatization:** Transforming tokens into their canonical (lemma) form, facilitating uniform semantic analysis.

6.2 EXPLORATORY TEXT ANALYSIS AND WORD CLOUDS

Exploratory Text Analysis (ETA) was conducted to reveal patterns, frequent terminologies, and general themes embedded in job notes. Word frequencies were computed, and word clouds were generated to visualize prominent terms and phrases effectively. The frequency $f(w)$ of each token w across all documents D is calculated as:

$$f(w) = \sum_{d \in D} \text{count}(w, d) \tag{12}$$

Figure 12 presents the top 20 most frequently occurring words found in the job notes cor-

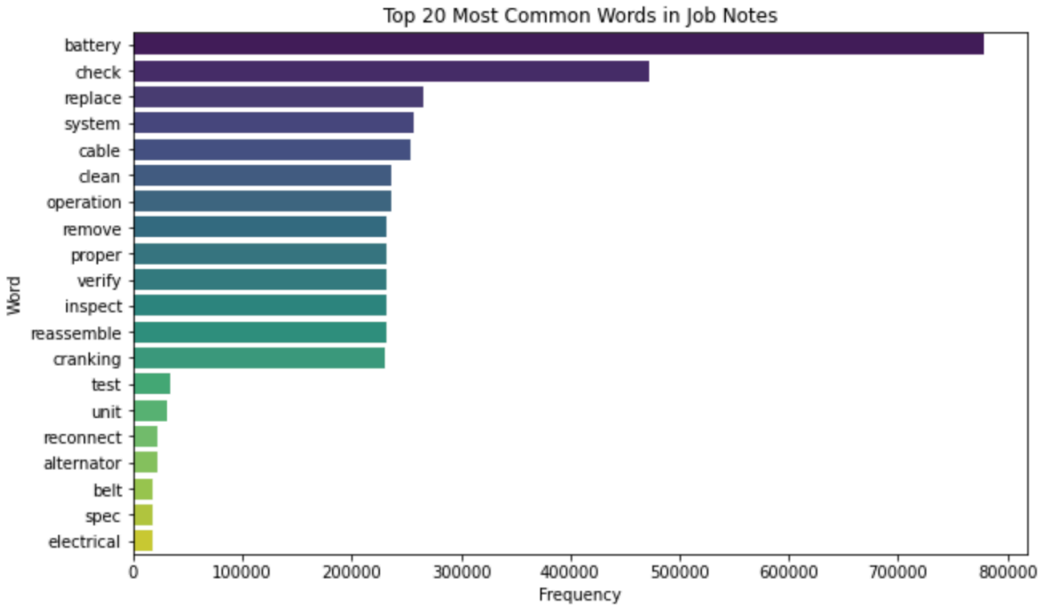


FIGURE 12: TOP 20 MOST COMMON WORDS IN JOB NOTES

pus, as measured by absolute frequency counts. The chart reveals that terms such as "battery," "check," "replace," "system," and "cable" overwhelmingly dominate the documentation, with "battery" alone appearing nearly 800,000 times. Other highly ranked terms include operational and maintenance actions like "clean," "operation," "remove," "proper," "verify," "inspect," and "reassemble," as well as specific components such as "cranking," "alternator," "belt," and "electrical." The concentration of these keywords highlights the primary focus areas within Penske's fleet maintenance operations—namely, battery-related diagnostics, component replacement, and system inspections. This frequency analysis confirms that preventive maintenance and electrical system servicing are central to day-to-day operational workflows, and it also provides a data-driven basis for optimizing training, resource allocation, and maintenance protocols around the most common and impactful issues.

Word clouds provided intuitive visual summaries where the font size of each token w is directly proportional to its frequency $f(w)$. This enabled rapid identification of recurring issues, terminologies, or operational themes. The word cloud depicted in Figure 13 provides a visual summary of the most frequently occurring terms and phrases extracted from maintenance job notes. Larger and bolder words, such as "verify," "replace," "battery," "cable," "remove," "proper," and "inspect," indicate higher frequencies and thus greater prominence within the job documentation. The central presence of terms related to battery servicing and electrical systems (e.g., "cranking," "system," "check," "reassemble," "clean," and "operation") highlights that battery maintenance and related electrical operations are dominant themes in Penske's fleet service activities. This visualization not only underscores recurring operational tasks but also helps to quickly identify core areas of focus and potential bottlenecks in maintenance workflows. By leveraging these insights, management can prioritize targeted training, resource allocation, and process improvements in the most commonly encountered maintenance domains, leading to enhanced service efficiency and reduced downtime.

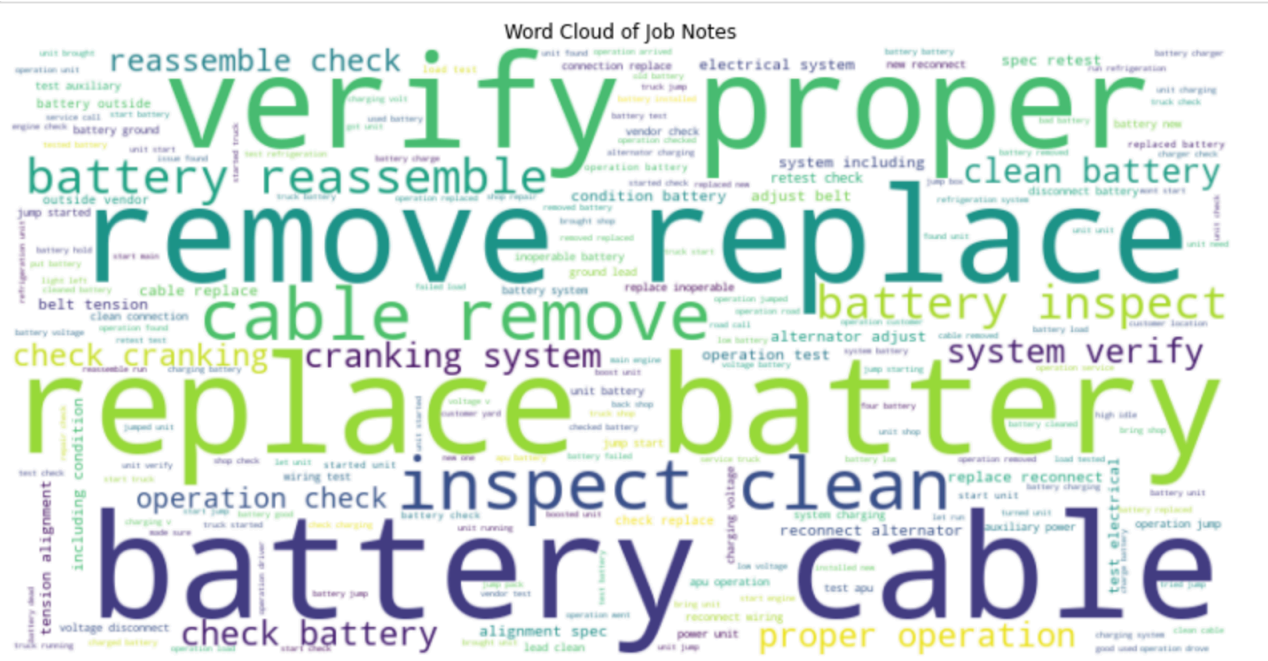


FIGURE 13: WORD CLOUD OF THE JOB NOTES

6.3 TOPIC MODELING AND KEYWORD EXTRACTION

Latent Dirichlet Allocation (LDA), an unsupervised generative probabilistic model, was employed to identify latent topics within the job notes corpus. Mathematically, the LDA model assumes that each document d is a distribution over topics θ_d, and each topic z is a distribution over words ϕ_z. The probability of observing a word w in document d is:

$$P(w|d) = \sum_{z=1}^{K} P(w|z)P(z|d) = \sum_{z=1}^{K} \phi_{zw}\theta_{dz} \tag{13}$$

where:

- K represents the number of latent topics,
- $\phi_{zw} = P(w|z)$ is the probability of word w given topic z,
- $\theta_{dz} = P(z|d)$ is the probability of topic z given document d.

After model training, we extracted and interpreted keywords associated with each latent topic. Below are the five topics we created based on the job notes:

Topic 1: Battery Testing and Replacement Procedures This topic centers on the technical procedures involved in battery testing and replacement. Frequently occurring words such as "battery," "test," "voltage," and "replaced" indicate that a substantial portion of maintenance operations involve systematic diagnostic testing and the subsequent replacement of faulty batteries or cables.

$$\begin{aligned} P(w|z_0) = &0.141\,\text{(battery)} + 0.028\,\text{(replaced)} + 0.023\,\text{(load)} + 0.021\,\text{(voltage)} \\ &+ 0.020\,\text{(test)} + 0.019\,\text{(found)} + 0.018\,\text{(cable)} + 0.018\,\text{(tested)} \\ &+ 0.017\,\text{(removed)} + 0.017\,\text{(checked)} \end{aligned}$$

Topic 2: Truck Battery Jump Start and Service: Terms like "unit," "jump," "start," and "truck" reveal that many job notes address emergency scenarios where vehicles require jump starts or urgent roadside assistance. The presence of words like "customer" and "shop" further points to direct customer impact and the need for efficient service recovery processes.

$$\begin{aligned} P(w|z_1) = &0.087\,\text{(unit)} + 0.052\,\text{(jump)} + 0.047\,\text{(start)} + 0.044\,\text{(truck)} \\ &+ 0.030\,\text{(started)} + 0.024\,\text{(shop)} + 0.014\,\text{(driver)} + 0.012\,\text{(customer)} \\ &+ 0.012\,\text{(battery)} + 0.011\,\text{(left)} \end{aligned}$$

Topic 3: Alternator and Electrical System Diagnostics: This topic is dominated by terms such as "alternator," "system," "belt," and "electrical," reflecting job notes focused on diagnosing and repairing broader electrical issues beyond the battery itself.

$$\begin{aligned} P(w|z_2) = &0.065\,\text{(alternator)} + 0.063\,\text{(system)} + 0.063\,\text{(belt)} + 0.062\,\text{(test)} \\ &+ 0.062\,\text{(spec)} + 0.062\,\text{(electrical)} + 0.062\,\text{(condition)} + 0.062\,\text{(replace)} \\ &+ 0.062\,\text{(cable)} + 0.061\,\text{(adjust)} \end{aligned}$$

Topic 4: Battery Inspection and Reassembly Process: Frequent use of words like "check," "inspect," "remove," "reassemble," and "verify" underscores a systematic approach to battery inspections and the careful reassembly of components post-maintenance.

$$\begin{aligned} P(w|z_3) = &0.188\,\text{(battery)} + 0.124\,\text{(check)} + 0.063\,\text{(remove)} + 0.063\,\text{(proper)} \\ &+ 0.063\,\text{(system)} + 0.063\,\text{(verify)} + 0.063\,\text{(cable)} + 0.063\,\text{(inspect)} \\ &+ 0.062\,\text{(replace)} + 0.062\,\text{(reassemble)} \end{aligned}$$

Topic 5: Battery and APU System Maintenance: This topic combines battery-related maintenance with references to the Auxiliary Power Unit (APU) and external vendors. The occurrence of terms like "vendor," "outside," and "APU" suggests that some battery servicing involves collaboration with third-party providers and maintenance of auxiliary systems.

$$\begin{aligned} P(w|z_4) = {} & 0.147\,\text{(battery)} + 0.089\,\text{(replace)} + 0.065\,\text{(test)} + 0.053\,\text{(check)} \\ & + 0.051\,\text{(outside)} + 0.051\,\text{(vendor)} + 0.036\,\text{(unit)} + 0.035\,\text{(apu)} \\ & + 0.034\,\text{(power)} + 0.032\,\text{(system)} \end{aligned}$$

Where $P(w|z)$ denotes the likelihood of observing word w given topic z. These probabilities reflect the relative significance and prevalence of words within each topic, providing a quantitative framework for interpreting thematic structures in the corpus of job notes.

6.4 INSIGHTS INTO OPERATIONAL ISSUES

The text mining process generated insights into recurring operational issues captured within job notes. Topics revealed frequent maintenance-related keywords, emphasizing specific problems such as battery issues, engine malfunctions, and tire conditions. This granular understanding of operational patterns enables proactive interventions and refined maintenance scheduling, potentially minimizing downtime and improving overall fleet reliability.

By systematically interpreting extracted topics and keywords, management can better align operational strategies, allocate resources efficiently, and develop targeted training programs for service technicians. The analytical insights thus derived provide a robust foundation for informed decision-making and continuous operational improvement.

7 CONCLUSION AND RECOMMENDATIONS

This collaborative analytics initiative between Penske and Saint Joseph's University has successfully demonstrated the strategic value of integrating predictive analytics into fleet maintenance operations. By employing advanced statistical and machine learning techniques, the project accurately predicted battery replacements, effectively forecasted repair timelines, identified anomalous replacement events, and extracted critical insights from unstructured technician job notes. The predictive models—Random Forest for battery replacements, LASSO regression for repair scheduling, and One-Class SVM for anomaly detection—provided robust, actionable insights, enabling a transition from reactive maintenance approaches to proactive, predictive strategies. Additionally, advanced text analytics using LDA clearly identified recurring operational themes from technician job notes, further enhancing targeted preventive maintenance practices.

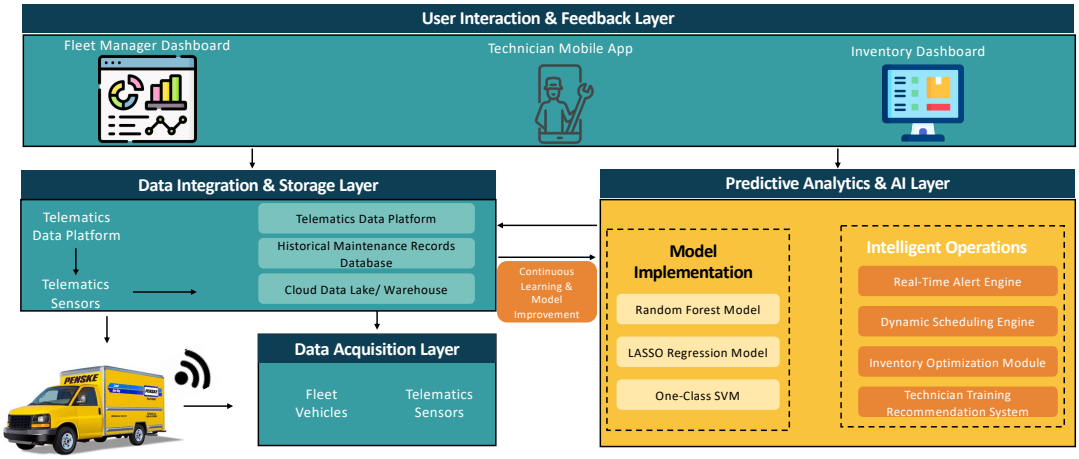


FIGURE 14: BUSINESS RECOMMENDATION IMPLEMENTATION ARCHITECTURE

Based on the analytical insights and predictive models developed, Penske is advised to implement the following strategic actions to enhance fleet reliability, optimize operational efficiency, and maintain competitive leadership in the fleet management industry, as Figure 14 shows:

- Deploy Real-Time Predictive Maintenance Alerts: Integrate the trained Random Forest classification model into existing fleet telematics platforms, providing real-time alerts when vehicles surpass critical thresholds such as high cumulative mileage (especially above 343,021 miles), significant duty cycle intensities, or abnormal cranking voltages. This proactive strategy will substantially reduce unexpected battery failures, costly downtime, and identify the unnessary battery replacement.

- Optimize Repair Scheduling with Predictive Insights: Leverage the LASSO regression model to refine maintenance scheduling by proactively identifying vehicles likely to experience delayed repairs. Specifically, prioritize inspection of vehicles exhibiting factors such as higher minimum odometer readings, moderate ambient temperatures, or subtle voltage irregularities. Conversely, vehicles demonstrating urgent telematics signals (e.g., extremely high odometer readings or pronounced cranking voltage issues) should be fast-tracked for immediate attention.

- Enhance Anomaly Detection and Prevent Unnecessary Replacements: Regularly utilize the One-Class SVM anomaly detection approach to systematically monitor fleet-wide telematics data and identify atypical battery replacement events. By detecting abnormal telematics patterns—such as significantly elevated odometer readings and declining cranking voltages occurring 5–8 weeks prior to replacement—this approach can flag potentially unnecessary replacements. Promptly alerting technicians to these anomalies enables careful verification of battery condition, ensuring replacements are truly war-

ranted. This targeted process prevents unnecessary maintenance costs, optimizes resource utilization, and minimizes unnecessary vehicle downtime.

- Refine Inventory Management and Operational Policies: Adjust inventory management systems to align replacement battery availability with predictive model forecasts, ensuring optimal stock levels where and when they are most needed. Revise operational policies based on data-driven thresholds derived from mileage, telematics indicators, and vehicle configurations (e.g., sleeper or liftgate-equipped vehicles) to proactively extend vehicle uptime and lifespan.
- Standardize and Strengthen Technician Training Programs: To collect more technician job notes data, incorporate insights derived from the topic modeling of job notes into technician training programs. Specifically, address identified key operational areas such as systematic battery testing, electrical system diagnostics, emergency jump-start procedures, and APU maintenance.
- Establish a Continuous Improvement Framework: Institutionalize an ongoing analytics program that continuously updates and refines predictive models with new telematics and maintenance data. Periodic model retraining and validation will sustain high predictive accuracy and relevance, ensuring Penske's maintenance operations remain adaptive and resilient to evolving fleet conditions.